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Paleomagnetic evidence for a diachronic clockwise rotation of the Coastal Range, eastern Taiwan

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ABSTRACT

Paleomagnetic results obtained from over 2100 cores sampled at 132 early Pliocene to late Pleistocene sites in the Coastal Range of eastern Taiwan indicate that, since the late Pliocene, the margin of the Philippine Sea plate has undergone a clockwise rotation of about 30° as a direct consequence of the Plio-Pleistocene collision of this plate with the Chinese Continental plate. The rotation is diachronic and started between 3 and 4 Ma ago in the northern Coastal Range then propagated southward at an average speed of the order of 70 ± 10 km/Ma. This value is in very good agreement with estimates of the southward propagation of the collision between the Philippine Sea plate and the Eurasian margin in Taiwan. It is suggested that the rotation reflects the deformation of the Philippine Sea plate at depth and does not results from the shallow deformations involved in the building of the Coastal Range orogen.

1. Introduction

The island of Taiwan is an active orogen resulting from the collision of the Philippine Sea plate with the continental Chinese plate [1–3], located between two subduction zones of opposite vergences. North of the island, the Philippine Sea plate is subducted northward beneath the Ryukyu arc, whilst in the south the oceanic part of the Eurasian plate, corresponding to the South China Sea, is subducting eastward beneath the Philippine Sea plate [4]. The present-day relative rate of rotation of the Philippine Sea plate with respect to Eurasia is $1.2 \pm 0.2°$/Ma, about a pole of rotation located north of the Japanese archipelagos [5–8]. In the region of Taiwan, this corresponds to a speed of convergence of about 70 km/Ma along the direction N120°E.

The island itself is composed of two main parts. The western part, which includes from west to east the coastal plain, the foothills and the Central Range, belongs to the deformed Chinese continental margin. The eastern part, the Coastal Range, results from the deformation of the Luzon arc located at the western boundary of the Philippine Sea plate. These two parts of the islands are separated by the Longitudinal Valley fault which is believed to be the boundary between the Philippine Sea and the Eurasian plates in the collision zone (Fig. 1A). This tectonic feature is described as an active thrust [9].

A kinematic model for the evolution of the Taiwan collision has been proposed by Suppe [10,11] on the basis of geological, geophysical and geodynamical studies. According to this author, the collision started about 4 Ma ago in the north of the island and then propagated southward with a mean rate of about 84 km/Ma. Recently Barrier [12] and Barrier and Angelier [13], on the basis of a slightly different orientation of the reference frame, have estimated the propagation of the collision at about 68 km/Ma.

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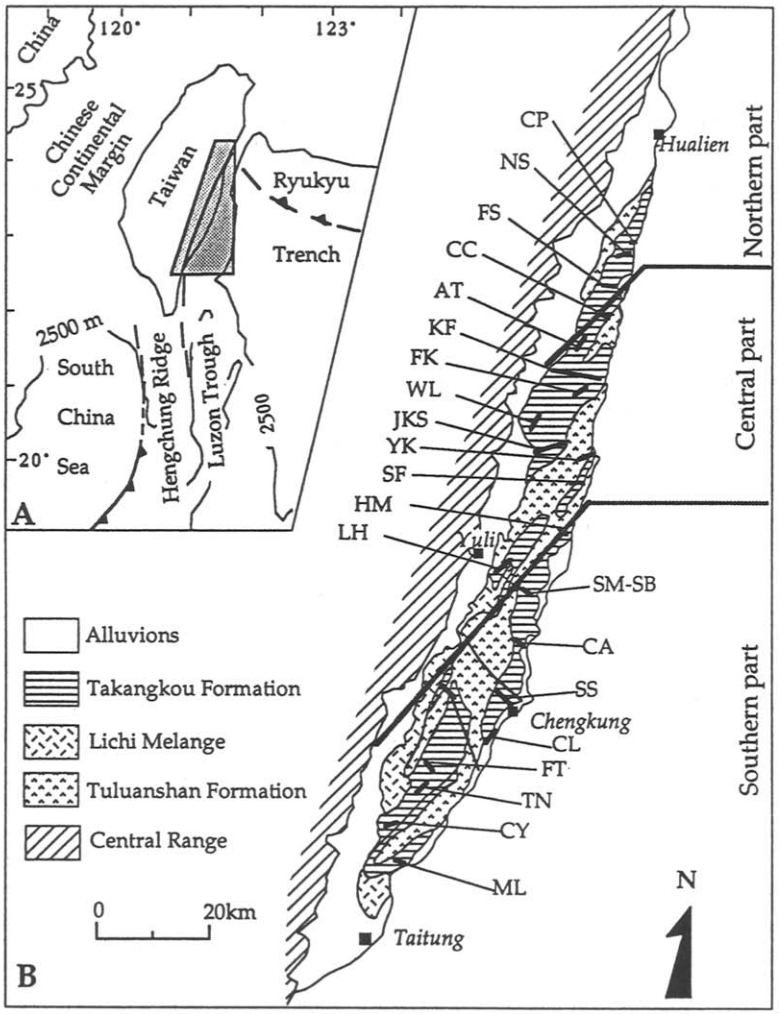


Fig. 1. A. Present-day configuration of Taiwan area. B. Schematic map of the Coastal Range showing the geographical distribution of the sections sampled for the paleomagnetic study after Lee et al. [15].

From a structural point of view, the Coastal Range is constituted by an "en échelon" fold system composed of numerous synclines and anticlines [14]. The strikes of the fold axes are distributed from N020°E, i.e. parallel to the overall trend of the chain and of the longitudinal valley, to N040°E which characterizes the "en échelon" fold system itself. This obliquity between the fold axes and the general trend could be due to local rotations about vertical axes related to the collision process. Were this the case, the angle of deviation is large enough to be detected by paleomagnetism, even in very recent formations.

Recently we have studied the anisotropy of magnetic susceptibility of over 2100 samples from 132 sedimentary sites sampled in the Coastal Range in order to reconstruct the evolution of the direction of the stress field in the Coastal range since the early Pliocene [15]. In this paper, the same sites are studied paleomagnetically in order to detect possible rotations of segments of the Philippine Sea plate and to define their time relation with the different phases of the collision proposed by the geodynamical models.

2. Geological setting and sampling

The geological characteristics of the Coastal Range of Taiwan have been described in a number of papers [14,16–18] and a description of the

paleomagnetic sampling has been given in Lee et al. [15] and in a very detailed way in Lee [19]. Thus only a short summary of the main geological features of this zone is given here.

The stratigraphic sequence of the Coastal Range has been first described by Hsu [14]. The basement is constituted by the Miocene volcanics of the Tuluanshan formation composed of tuffs, tuffaceous sandstones, pillow lavas and andesitic agglomerates. The transition between the basement and the sedimentary cover is marked in some areas by reef limestones dated at the Miocene–Pliocene boundary age, as determined by nannofossil analysis.

The sedimentary cover of the Coastal Range, which is called the Takangkou Formation, conformably overlies the Tuluanshan Formation. Its age ranges from early Pliocene to early Pleistocene [16,17]. Its lower part is composed of volcanogenic bioclastic turbidites dominated by thin alternating sandstones and shales. The upper part consists of various facies of turbidites which end up with coarse-grained levels and even conglomerates. The lower Pleistocene turbiditic formation which has been described on the top of the Takangkou Formation in the Hsiukuluanchi valley is called the Chimei Formation. Finally, in the south of the Coastal Range, the Lichi Formation is intercalated within the lower part of the Takangkou Formation. This unit, of chaotic nature, is not suitable for paleomagnetic studies.

In the "en échelon" fold system, the anticlines are marked by elongated outcrops of the Tuluanshan Formation while the upper part of the Takangkou Formation and the Chimei Formation exist in the synclines. The anticlines are generally west-vergent faulted overfolds. The eastern limbs are thus better developed than the western ones. The main structures of the Coastal Range result from the compressive phase related to the collision and which is active since the late Pliocene. According to the structural analyses, the present-day direction of compression is N120°E [12].

A total of 132 sedimentary sites (more than 2100 cores) were sampled all along the Coastal Range in the blue grey marls of the Takangkou and the Chimei Formations. We have preferentially sampled the fine-grained marls because it indicates that the deposition of the sediments occurred in a calm water. The sections too close to major faults or containing synsedimentary structures were avoided. The sampling sites were geographically distributed in various sections, in the different valleys of the Coastal Range (Fig. 1B, and detailed description in [19]).

Sampling was done with an electric drill. The corer is a 25-mm barrel with a sintered diamond cutting head cooled by water or compressed air. Orientation of the cores was by standard paleomagnetic techniques using both magnetic and sun compasses. Each core gave 3–5 specimens, 22 mm long.

3. Magnetic mineralogy and demagnetization

The analysis of the anisotropy of the low field magnetic susceptibility has already been reported in a previous paper [15]. It has been observed that in about 20% of the sites the magnetic fabric is planar, with K_{min} perfectly perpendicular to the bedding. In another 60%, a magnetic lineation, reveals a tectonic contribution to the fabric. However as K_{min} remains perpendicular to the bedding, the sedimentary fabric still dominates. Finally in the remainder of the sites, the susceptibility ellipsoid is prolate with K_{min} and K_{int} virtually identical and forming a girdle about K_{max}. This fabric yields evidence for a dominant tectonic fabric, although visually all the sites appear undisturbed and only tilted. Also we did not find any correlation of the preferred occurrence of these typical forms of the susceptibility ellipsoid with either the geographic or the structural position of the sites. As discussed below, the direction of the remanent magnetization at the different sites does not appear to be significantly affected by the tectonic contribution revealed by the analysis of the magnetic fabric.

The magnetic mineralogy of one to four cores per site has been carefully investigated, either on whole rock or using magnetic extracts obtained with a method similar to the one of Papamarinopoulos et al. [20].

Saturation of the isothermal remanent magnetization (IRM) is systematically attained below 0.2 T and the value of the median destructive field of the saturation IRM is less than 30 mT [15,19], suggesting that magnetite is the main magnetic carrier. This is confirmed by the Curie balance analysis which shows that the induced magnetiza-

tion progressively decreases and finally disappears around 580°C [15,19]. A small change in the slope of the heating curve is, however, often observed around 330–360°C and is correlated with an increase in the value of the low field magnetic susceptibility and with an abrupt change in the remanent magnetization (Fig. 2A). Also the heating and cooling curves of the thermomagnetic analysis are usually not reversible [15,19] despite the nitrogen atmosphere, suggesting that some transformation of part of the magnetic minerals occurs during the heating. The thermal instability above 350°C could suggest that the unheated samples may be dominated by maghemite, which is characterized by the same coercivities as magnetite and converted to hematite during the heating. However, the X-ray diffraction analysis of unheated magnetic extracts shows very clearly the presence of magnetite as the main magnetic mineral (Fig. 2B). Grain sizes are in the pseudo-single domain range [15,19], as determined by the method of King et al. [21], with the larger sizes corresponding to sites clearly in a more proximal situation with respect to the source of detrital material.

In summary, these results show that the magnetic mineralogy is dominated by magnetite with variable amounts of thermally less stable magnetic minerals. Thermal demagnetization of pilot samples indicated that these instabilities could eventu-

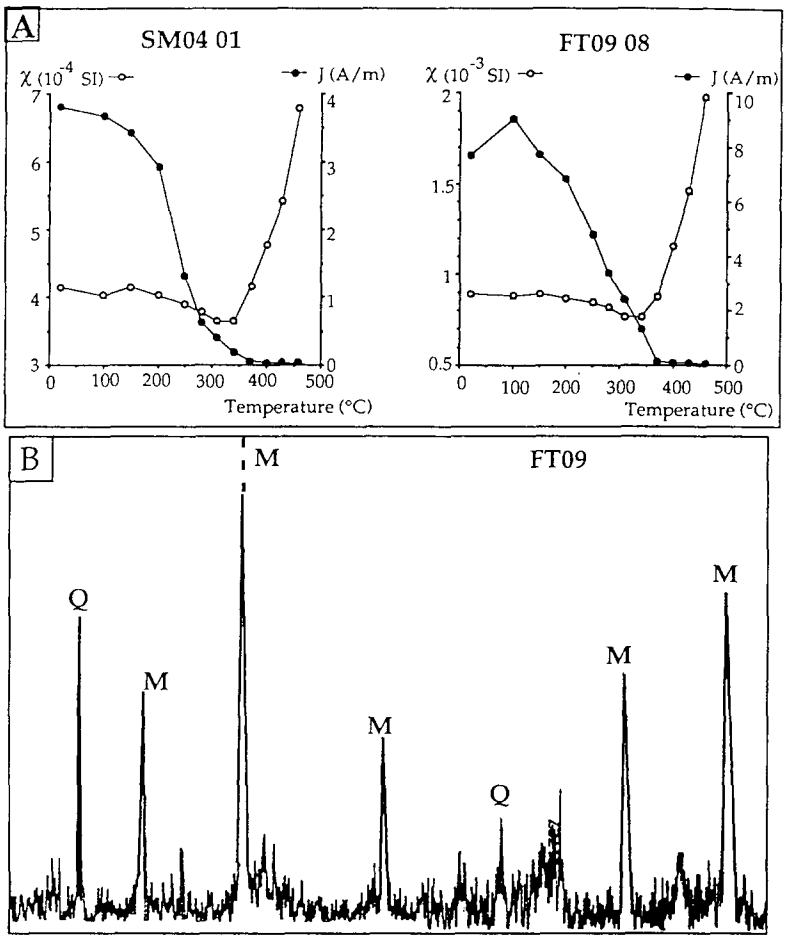


Fig. 2. A. Evolution of the intensity and of the low field susceptibility values of two samples during thermal demagnetization indicating that in these cases, part of the magnetic minerals has been transformed around 350°C. B. X-ray spectrum of an unheated magnetic extract from one of the two sites shown in A indicating, in addition to the quartz (*Q*), the presence of magnetite (*M*) as the dominant magnetic carrier.

ally make quite delicate the interpretation of the demagnetization diagrams, some of which were characterized by an erratic behaviour after the 350°C step. Unfortunately, the AF demagnetizing technique also gave unsatisfactory results, because it proved unable to remove a viscous component present in all samples. On the other hand, this component was easily demagnetized by heating at about 250°C. Because of the very large number of samples considered in this study, we chose a stepwise demagnetization procedure which could be used almost routinely for all of them: first, the samples were thermally demagnetized up to 330°C (7 steps) in order to completely remove the viscous component, then a stepwise AF treatment was used up to 60 mT in 6 to 7 steps. Typical examples of orthogonal plots obtained by this procedure are shown in Fig. 3. In the very large majority of the cases, the orthogonal diagrams obtained with this method show a perfectly linear trend decreasing to the origin. We believe that the component isolated in this way represent the primary stable component of the magnetization. This interpretation is supported by the fact that when samples of both polarities are present in the same site, their paleomagnetic directions are exactly opposed which is a good indication of good stability and satisfactory cleaning. In a general way, it proved impossible to sample both flanks of the structures because the western flanks of the anticlines and the eastern flanks of the synclines are not well developed. However, in the three cases where it

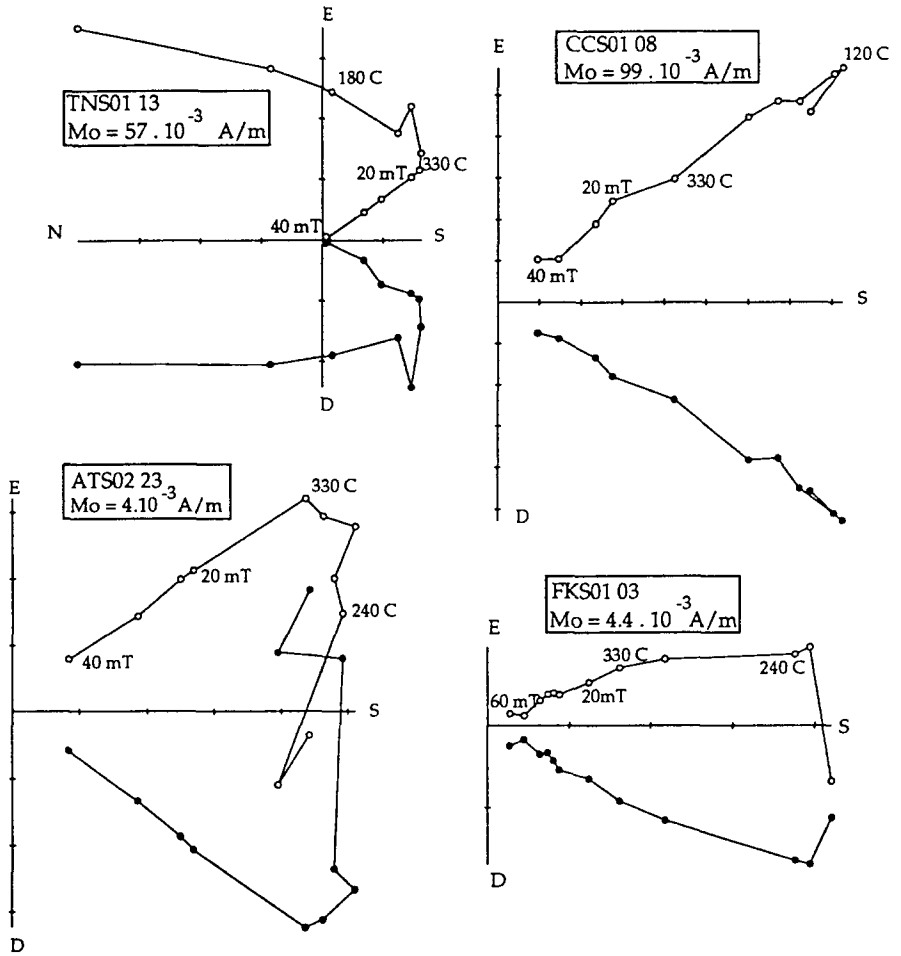


Fig. 3. Typical demagnetization diagrams obtained using first thermal demagnetization up to 330°C and then AF demagnetization. Open and filled circles correspond to the vertical and horizontal components respectively.

TABLE 1

Mean paleomagnetic directions calculated at each site from the Coastal Range of Taiwan and reported in declination and inclination before and after bedding correction (b.c). The angles of relative rotation of the Coastal Range with respect to the Philippine Sea Plate have also been reported. For each site, the biostratigraphic age is given using the number of the Nannofossil Zone (L = Lower; M = Middle; U = Upper). The absolute age was determined by comparison between bio- and magnetostratigraphc age of the sites (see text). The bedding attitude is given as the strike of the horizontal line, the dip of the plane, the letter indicating the approximate direction of dip.

Site	n	Before b.c.		After b.c.		α_{95} (°)	k	Relative rotation	Age		Bedding attitude
		D (°)	I (°)	D (°)	I (°)				biostr.	absolute (Ma)	
Northern region											
"Old sites"											
FS 01	7	19.5	27.0	60.0	28.1	12.3	18.7	53.6	12	5.35	033, 68E
FS 02	8	225.0	−21.8	236.8	−23.3	15.2	14.3	50.7	12	5.35–4.77	054, 28S
FS 03	6	351.7	15.0	20.1	34.5	11.0	38.0	14.5	12	4.77–4.57	025, 51E
FS 04	8	179.2	−14.7	224.9	−30.2	8.2	46.0	39.7	12–13	4.57–3.97	030, 87E
NS 01	8	221.4	−83.7	190.0	−23.8	13.1	18.9	5.5	14	3.88–3.60	096, 61N
NS 02	8	231.6	−34.2	210.6	−29.5	8.1	47.2	26.1	14	3.88–3.60	049, 33N
NS 03	8	237.3	−39.5	223.5	−32.6	13.4	18.0	39.0	14	3.88–3.60	070, 20N
"Young site"											
CP 02	7	201.0	−52.8	184.4	−31.0	3.6	288.0	1.6	17	2.40–2.20	060, 28N
Central region											
"Old sites"											
LH 01	8	182.2	−0.9	207.6	−28.3	10.0	31.9	22.5	13–15	4.40–3.97	031, 83E
LH 02	8	8.1	−0.9	28.2	23.6	12.2	21.6	23.5	13–15	3.97–3.88	032, 83E
LH 03	6	182.4	−4.0	204.1	−28.8	16.3	17.8	20.0	15	3.60–3.40	032, 68E
LH 05	8	184.6	−7.0	215.2	−30.3	8.5	43.8	31.0	15	3.60–3.40	032, 78E
HM 01	8	0.8	30.1	19.2	26.6	4.8	134.1	14.5	14–15	3.97–3.88	004, 34E
HM 02	8	10.0	18.0	23.2	22.6	5.2	116.6	18.5	14–15	3.97–3.88	024, 35E
HM 03	8	185.7	5.0	198.0	−30.7	10.3	29.7	13.6	15	3.88–3.40	044, 59E
HM 04	8	9.5	17.6	21.2	29.1	5.7	97.1	17.6	16	3.40–2.47	036, 29E
CC 01	8	189.4	−22.6	220.7	−21.7	8.3	45.1	36.3	14–15	3.88–3.40	022, 67E
YK 02	12	178.2	−2.7	195.7	−26.4	6.4	47.0	11.5	15	3.60–3.40	028, 63E
YK 03	14	203.6	−10.6	225.6	−20.8	6.5	38.2	41.4	15	3.60–3.40	055, 55S
YK 04	14	185.6	−6.8	214.5	−30.2	6.2	41.7	30.3	15	3.60–3.40	032, 61E
YK 05	21	186.7	14.2	195.3	−26.2	8.1	16.5	11.1	15	3.60–3.40	035, 73E
TW 09	8	179.8	23.0	181.3	−22.6	14.8	14.9	−3.0	15	3.60–3.40	031, 73E
TW 03	8	0.2	34.8	45.3	34.6	3.6	240.5	41.7	16–18	3.40–2.47	019, 58E
TW 04	8	12.0	26.7	20.1	17.3	5.7	94.9	16.5	16–18	3.40–2.47	164, 23E
SF 01	10	356.2	22.8	18.0	38.9	8.0	37.4	14.4	16	3.40–2.47	031, 38E
SF 02	10	358.6	13.2	21.2	36.8	8.0	37.4	17.6	16	3.40–2.47	036, 50E
SF 03	10	1.5	8.7	11.5	18.3	7.0	48.6	7.9	16	3.40–2.47	020, 40E
SF 04	10	2.8	9.6	14.8	18.8	7.0	48.6	11.2	16	3.40–2.47	020, 45E
SF 05	10	359.2	24.4	25.1	38.2	4.0	146.8	21.5	16	3.40–2.47	030, 42E
KF 05	9	16.5	4.0	23.9	19.8	8.0	37.0	20.3	16	3.40–2.47	040, 68E
KF 06	7	191.3	9.9	194.8	−21.9	11.4	29.0	12.1	16–18	2.47–1.88	058, 44E
KF 07	10	199.0	3.2	213.5	−32.5	9.0	29.8	30.8	16–18	2.47–1.88	060, 60S
KF 08	8	4.2	5.0	11.7	26.9	7.0	63.6	8.1	16	3.40–2.47	049, 33S
KF 09	10	195.0	−41.4	181.1	−26.4	10.0	24.3	−2.0	16–18	2.47–1.88	044, 25N
FK 01	8	201.3	9.4	206.8	−24.0	7.3	57.9	24.2	16–18	2.47–1.88	063, 51S
FK 02	7	197.7	25.5	193.1	−19.8	4.9	150.0	10.5	16–18	2.47–1.88	043, 84E
FK 03	7	219.3	27.3	218.9	−25.1	10.0	37.1	36.2	16–18	2.47–1.88	098, 57S

TABLE 1 (continued)

Site	n	Before b.c.		After b.c.		α_{95} (°)	k	Relative rotation	Age		Bedding attitude
		D (°)	I (°)	D (°)	I (°)				biostr.	absolute (Ma)	
"Young sites"											
CC 03	7	21.4	37.8	1.9	16.0	11.8	27.1	0.0	L-19	1.88–1.66	039, 42W
CC 04	8	39.8	11.8	24.7	18.1	9.9	32.4	23.5	M-19	0.98–0.91	029, 65W
AT 01	5	185.6	−21.2	194.2	−7.7	4.7	268.2	12.7	19	1.66–0.73	325, 41E
AT 02	5	199.6	11.5	211.3	−25.0	8.8	50.0	30.0	M-19	1.30–0.98	059, 59S
KF 01	10	198.3	−40.6	200.4	−37.6	6.0	65.8	19.0	M-19	1.30–0.98	024, 08E
KF 02	10	359.0	20.7	354.6	13.2	5.0	94.3	−7.0	L-19	1.88–1.66	025, 16W
KF 03	10	8.1	28.6	3.8	23.0	6.0	65.8	1.6	L-19	1.88–1.66	050, 10N
KF 04	10	188.2	−62.3	198.2	−48.8	9.0	29.8	16.8	M-19	1.30–0.98	130, 15N
WL 01	8	194.6	−28.0	200.0	−7.8	7.7	52.4	18.4	19	1.66–0.98	143, 29E
WL 02	8	188.3	−22.5	199.0	−13.6	8.7	41.9	17.4	19	1.66–0.98	176, 33E
WL 04	6	18.2	33.0	20.0	12.0	16.9	16.6	17.8	19	1.88–1.66	060, 28E
JK 03	9	130.0	−25.0	168.6	−46.0	7.0	55.1	−13.0	L-19	1.66–1.30	170, 53E
JK 04	8	153.0	3.2	160.5	−20.3	7.0	63.6	−21.0	M-19	1.30–0.98	002, 52E
JK 05	8	164.3	−8.8	181.2	−29.0	10.0	31.4	0.0	M-19	1.30–0.98	015, 50E
JK 06	9	149.0	−9.4	172.0	−38.1	14.0	14.5	−9.0	M-19	1.30–0.98	010, 55E
JK 07	8	181.4	−3.0	185.0	−10.7	9.0	38.8	3.6	M-19	1.30–0.98	018, 30E
JK 08	9	166.3	−12.8	186.9	−32.2	8.0	42.4	5.5	M-19	1.30–0.98	018, 50E
JK 0B	8	172.0	−8.0	182.8	−18.2	11.0	26.3	1.4	M-19	1.30–0.98	010, 45E
JK 0C	5	154.1	−4.3	158.0	−24.5	10.4	36.3	−23.0	M-19	1.30–0.98	005, 50E
JK 0D	9	357.8	21.2	22.2	19.6	6.1	58.7	21.0	M-19	0.98	005, 50E
JK 0E	16	157.3	−2.7	165.9	−20.2	8.0	22.2	−15.0	M-19	0.91–0.73	004, 44E
JK 0F	12	170.0	−20.0	179.9	−25.3	5.0	76.3	−1.0	U-19	0.91–0.73	172, 24E
JK 0G	12	162.0	−24.4	176.0	−19.0	7.6	27.7	−5.0	U-19	0.91–0.73	128, 17N
JK 0H	7	354.5	28.8	0.3	25.0	8.4	36.0	0.0	U-19	< 0.73	040, 10E
TW 10	10	163.7	−5.2	166.0	−14.5	7.6	41.6	−13.0	U-19	0.91–0.73	011, 31E
TW 11	12	163.0	−69.3	179.6	−29.0	9.3	18.7	−1.0	U-19	0.91–0.73	variable
Southern region											
"Old sites"											
SB 01	8	181.0	−35.8	212.1	−36.3	8.6	42.7	30.6	19	1.66–0.73	017, 41E
SM 01	8	197.3	7.0	207.5	−18.7	7.4	57.2	24.8	16–18	2.47–1.88	037, 84E
SM 02	7	185.4	−4.0	210.1	−23.8	9.0	45.7	27.4	16–18	2.47–1.88	029, 83E
SM 03	8	187.7	−20.8	202.8	−19.0	8.4	44.9	20.1	16–18	2.47–1.88	013, 40E
SM 04	8	179.4	−42.8	206.0	−41.5	8.1	48.0	23.4	16–18	2.47–1.88	010, 29E
CA 01	8	205.0	−11.7	211.7	−7.6	5.0	121.7	30.1	M-19	1.66–0.98	021, 39E
CA 02	8	194.7	−42.2	203.0	−29.4	5.3	119.4	21.4	M-19	1.66–0.98	149, 17E
CA 03	8	205.2	−45.0	213.1	−25.6	5.8	92.4	31.6	19	1.66–0.73	147, 22E
CA 04	8	198.0	−27.0	217.3	−24.5	8.0	48.8	34.6	16–18	2.47–1.88	010, 51E
CA 05	8	14.0	46.6	46.4	42.5	8.2	46.2	44.2	L-19	1.88–1.66	017, 29E
CA 06	8	12.1	20.3	25.4	23.4	8.0	49.4	23.2	L-19	1.88–1.66	022, 32E
TW 07	10	7.0	1.2	22.0	35.7	6.9	49.7	18.0	15–16	3.40–3.18	041, 55E
SS 01	8	186.3	−13.0	211.8	−25.8	10.8	27.4	30.0	L-19	1.66–1.30	029, 65E
SS 02	8	195.0	−34.0	203.1	−50.8	8.3	45.5	21.3	L-19	1.66–1.30	081, 19S
SS 03	7	5.7	24.6	34.0	29.8	12.1	25.9	30.4	16	3.40–2.47	023, 51E
CL 04	8	171.0	−12.8	185.1	−26.6	11.1	25.7	3.3	L-19	1.66–1.30	015, 40E
FT 03	6	355.3	16.7	31.5	21.9	9.3	52.6	27.5	15–16	3.40–3.18	016, 83E
FT 04	7	194.3	−5.7	200.5	−32.7	7.8	61.6	18.7	L-19	1.66–1.30	074, 32S
FT 05	8	216.0	−7.3	224.2	−22.5	8.0	49.1	42.4	L-19	1.66–1.30	066, 34S
FT 06	8	205.4	7.0	209.4	−27.3	5.7	97.2	27.6	L-19	1.66–1.30	085, 40S
FT 07	8	195.1	−29.1	221.8	−35.0	7.3	59.2	40.0	L-19	1.66–1.30	036, 41E
FT 08	7	183.8	−12.5	199.1	−29.7	13.6	20.7	17.3	L-19	1.66–1.30	035, 41E
FT 09	8	198.2	−19.2	213.8	−26.8	7.8	51.1	31.1	17–18	2.40–2.00	037, 36E
FT 10	7	200.0	−3.1	204.9	−8.0	6.4	89.9	21.2	M-16	3.18–2.92	028, 48E

TABLE 1 (continued)

Site	n	Before b.c.		After b.c.		α_{95} (°)	k	Relative rotation	Age		Bedding attitude
		D (°)	I (°)	D (°)	I (°)				biostr.	absolute (Ma)	
"Old sites"											
FT 11	7	1.3	23.6	39.8	32.1	9.0	46.2	35.8	L-16	3.40–3.18	027, 64E
FT 12	7	195.7	−12.7	203.0	−19.2	6.8	80.2	21.2	L-19	1.66–1.30	023, 43E
FT 13	8	198.0	16.5	197.4	−18.5	8.7	41.3	15.6	L-19	1.66–1.30	144, 44W
FT 16	8	12.5	41.5	55.8	54.1	11.8	23.0	51.8	15–16	3.40–3.18	070, 46S
FT 17	8	188.2	6.0	203.2	−26.1	9.4	35.4	21.4	L-19	1.66–1.30	039, 70E
FT 19	6	205.5	−5.4	211.4	−14.3	11.2	36.7	29.6	L-19	1.66–1.30	041, 34E
FT 21	8	248.0	−0.5	219.2	−43.1	9.8	32.9	35.0	15	3.60–3.40	020, 72W
CY 04	8	8.0	15.0	32.8	21.3	5.9	88.3	29.5	16–18	2.92–2.47	023, 67E
CY 05	7	200.4	−18.8	218.5	−18.4	4.8	157.2	34.8	16–18	3.18–2.92	027, 50E
ML 01	8	29.0	54.5	46.9	25.6	10.9	26.6	42.1	14–15	3.97–3.88	162, 35E
ML 02	8	211.5	−0.9	219.8	−16.7	6.7	68.6	35.6	M-15	3.60–3.40	050, 52S
"Young sites"											
FT 20	6	255.0	−67.2	199.7	−33.7	15.6	19.3	18.3	M-19	1.30–0.98	080, 50N
TN 01	8	243.3	−73.2	186.6	−32.3	11.1	25.9	5.2	M-19	1.30–0.98	077, 52N
TN 02	8	197.0	−67.1	177.1	−26.0	7.3	57.9	−4.0	M-19	1.30–0.98	073, 44N
TN 03	6	224.0	−66.2	185.5	−36.5	8.5	63.7	4.1	M-19	1.30–0.98	068, 40N
TN 04	8	242.7	−63.0	192.3	−38.0	10.0	31.9	11.0	M-19	1.30–0.98	065, 44N
TN 05	7	245.6	−64.6	194.5	−31.0	10.0	27.0	13.1	M-19	1.30–0.98	073, 45N
TN 06	8	255.1	−67.1	200.1	−34.6	6.8	66.7	18.7	M-19	1.30–0.98	080, 49N
CL 01	7	193.5	−7.6	196.9	−21.5	10.6	33.6	16.0	U-19	0.91–0.73	060, 19S
CL 02	6	171.0	−4.4	178.7	−24.5	8.9	57.1	−2.0	U-19	0.91–0.73	027, 35E

n: number of samples; α_{95} and k are the parameters of Fisher's statistic.

proved possible to sample both sides of the structures, the fold test proposed by McFadden and Jones [22] was positive at the 5% significance level proving that the component isolated by demagnetization predates the folding (Tables 1 and 2). For the other sites, one observes that the scattering of the paleomagnetic directions between sites of the same age and from the same structure is significantly smaller after tectonic correction (at the 1% significance level) even if they have been sampled in the same flank.

All these laboratory analyses and the complementary tests yield evidences that the primary component was successfully isolated and that the

TABLE 2

Positive fold test obtained from three structures in the Coastal Range using the method given by McFadden and Jones [22]

Sites	K_1/K_2	F (5%)	R_a	R_b	R_1	R_2	AR_1	AR_2	B
NS 03–FS 02	1.25	2.5	7.61	7.51	14.80	14.99	0.27	0.14	0.24
FT 08–FT 13	1.98	2.64	6.71	7.83	13.97	14.47	1.21	0.15	0.26
FT 11–FT 21	1.40	2.55	6.87	7.78	12.14	14.59	6.72	0.19	0.26

K_1 and K_2 are the precision parameters given by Fisher's statistic for each site (see Table 1). The ratio K_1/K_2 is compared to the value of the test F (5% significance level) with N_a and N_b degrees of freedom, N_a and N_b being the number of directions in each site. This first test permits to check that the two sites are characterized by the same dispersion.

R_a and R_b are the length of the resultant vector for each site. R_1 and R_2 are the length of the resultant vector of the two sites put together before and after bedding correction. The first member of the eq. (7) of McFadden and Jones [22] is calculated before (AR_1) and after (AR_2) bedding correction. B is the second term of the same equation at 5% level of significance. The three fold tests are positive after bedding correction ($AR_2 < B$) indicating that the two limbs are characterized by a pre-folding common true mean direction.

investigated formations are suitable for a reliable paleomagnetic study.

4. Paleomagnetic results

The precise ages of the sampling sites were obtained by biostratigraphic dating based on nannofossil associations. The obtained ages were then combined with the analysis of the magnetic polarities [19] which were obtained in each section using the magnetostratigraphic scale proposed by Berggren et al. [23] as the reference. This study allowed us to precisely define the age of some of the sites within a Nannofossil Zone.

Only a few outcrops of early Pliocene age suitable for a paleomagnetic study could be located. The rest of the sites are evenly distributed in age from middle Pliocene to late Pleistocene (Brunhes period). The stratigraphic position of the investigated sites is indicated in Table 1 along with the paleomagnetic results.

The mean paleomagnetic directions reported in Table 1 have been calculated using Fisher's statistics at each site [24]. Because our previous study [15] has revealed that a magnetic fabric of tectonic origin characterizes a majority of the sites, we have checked that this has not significantly deflected the paleomagnetic directions. For instance, sites WL 02, JK 03, 07, 08, 0C, 0G, 0H where a magnetic lineation of tectonic origin is observed, yield a mean paleomagnetic direction ($D = 359.2°$, $I = 23.7°$; $\alpha_{95} = 9.8°$) which is virtually identical to that ($D = 359.5°$, $I = 23.7°$; $\alpha_{95} = 9.7°$) obtained from sites AT 01, JK 04, 05, 06, 0D, 0F, TW 10, 11 of the same age characterized by a sedimentary fabric. Thus, the directions of remanent magnetization have not been affected by any tectonic effect.

Inclination values are somewhat scattered around a mean value of about 30°. This value is shallower than expected on the basis of a geocentered dipole field at the present-day latitude of Taiwan (40°). Because this shallowing is present also in the most recent sediments, it does not appear reasonable to ascribe it to a possible northward drift. Rather, it could reflect an inclination error such as that already observed elsewhere in the same kind of sediments and attributed to compaction [25,26].

The values of the declinations vary from 5° to 40°. It can be observed from Table 1 that the values of the declinations depend on both the geographical location and the age of the sampling sites.

5. Discussion

According to the study of slip vectors of the earthquakes which occurred at the plate boundary, compared to the regional kinematic data, the Philippine Sea plate has undergone a clockwise rotation at an average rate of about 1.2°/Ma over the last 4 Ma [5–8]. The value of the overall rotation of this plate calculated at each site according to its age and assuming a uniform rotation, has to be subtracted from the measured value at each site to interpret the paleomagnetic results in terms of deformation of the Coastal Range with respect of the stable part of the Philippine Sea plate. The value of the rotation corrected this way, called "relative rotation", is also given in Table 1.

To assess the tectonic significance of the paleomagnetic results we have considered separately three different regions of the Coastal Range according to their major structures, namely the northern, central and southern regions. These regions are indicated in Fig. 1 and the results corresponding to each of them identified in Table 1.

An examination of the data in this table shows that in each of the three regions, the results can be separated into two groups, depending on the age of the sites. A first group, comprising the oldest sites, is characterized in each region by consistent clockwise rotations of about 25°. On the contrary, a second group, comprising the younger sites, has not undergone any significant rotation. Although in a very few cases, the inter-site scatter of the data has introduced a small degree of uncertainty in assigning sites of intermediate age to one or the other of the two groups, this separation is overall quite obvious, as discussed below.

In the northern region, for instance, seven sites of age comprised between 5 and 3.7 Ma indicate a clockwise rotation of average value $33 \pm 12°$, whilst no rotation is observed from a single site dated at 2.3 Ma, as shown in Fig. 4A. It appears that for the older sites there is a distribution of the angles of rotation. There is however no direct relation between the age of the site and the angle of rotation, so that this distribution does not cor-

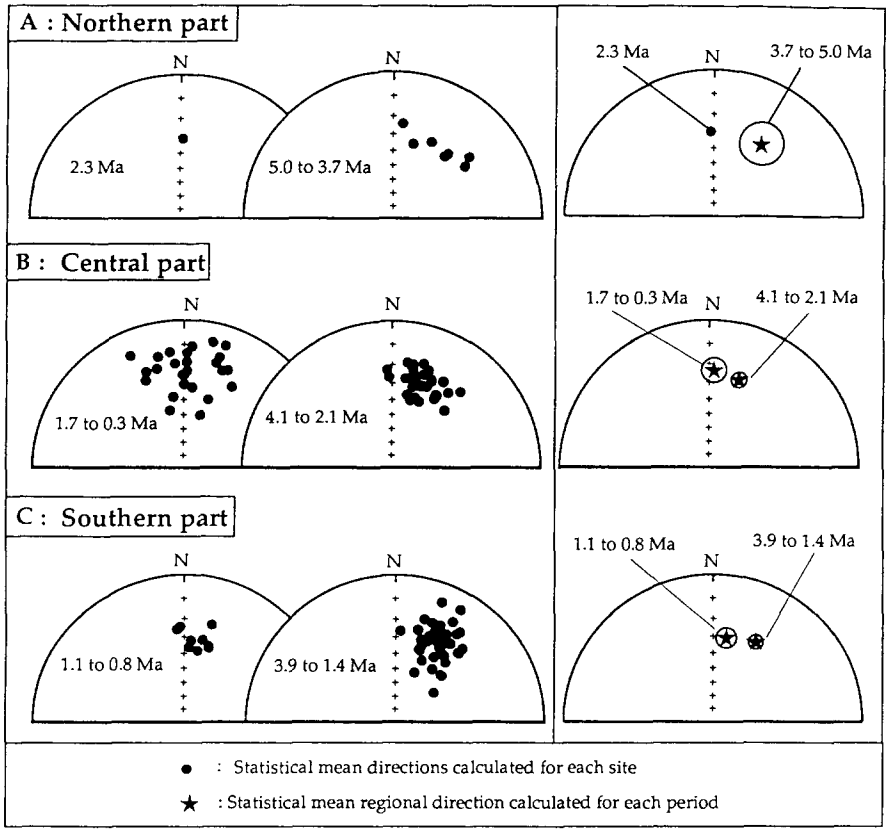


Fig. 4. Stereographic projection of the results obtained which have been separated in three groups corresponding to three different regions of the Coastal Range. For a better visual impression, all the reverse directions have been inverted through the origin.

respond to a rotation occurring progressively in time. We have no clear explanation for this streaked distribution which could reflect very local rotations about the mean value given by Fisher's statistics. These results suggests that the rotation occurred in this area during the late Pliocene and was probably completed by 2.3 Ma. However, because only one single site is available for the younger period, no reliable conclusion can be reached on the precise timing of the rotation.

On the other hand, many data are available in the central part of the Coastal Range. Twenty-nine sites, of normal and reverse polarities and ages ranging between 4.1 and 2.1 Ma, when averaged together yield a mean rotation of $19 \pm 4.2°$. No significant rotation ($4.3 \pm 6.5°$) was on the other hand indicated from averaging the 26 sites from the same region, of age comprised between 1.7 and 0.3 Ma. This suggests that the rotation has occurred in the central region of the Coastal Range between 2.1 and 1.7 Ma.

In the southern part, the rotation appears to be younger. Indeed, the thirty-five sites dated between 4 and 1.4 Ma have recorded a clockwise rotation of $27.5 \pm 4.4°$ whilst the younger nine sites show a much smaller angle of rotation ($8.7 \pm 6.2°$) (Fig. 4C).

Examination of the stereographic projections in Fig. 4, could suggest that our choice of time windows separating "rotated old" sites from "unrotated younger" sites was an arbitrary decision. However, the apparent scatter of the directions in each group, visible on these projections, is not related to the age of the site. Figure 5 shows the results for the central and southern parts. Here, the directions from sites having the same age have been averaged and plotted as a function of the age. The two groups are clearly separated and the

time windows well defined (in the northern part, the low number of sites does not permit us to construct the same diagram).

Three main conclusions can be drawn about the tectonic evolution of the Coastal Range. First, both in the central and in the southern parts, the rotation appears to have been very rapid (Fig. 5). Indeed, in both regions, the data suggest that the rotation of about 20–25° was completed in about half a million years (Figs. 5 and 6). Second, the data suggest that the rotation, the amount of which is not related to the strike of the structures, was not synchronous over the geographical extent of the Coastal Range. This diachronism from north to south of the Coastal Range is schematically presented in Fig. 6. Although the data are not precise in the northern part, we could draw a line

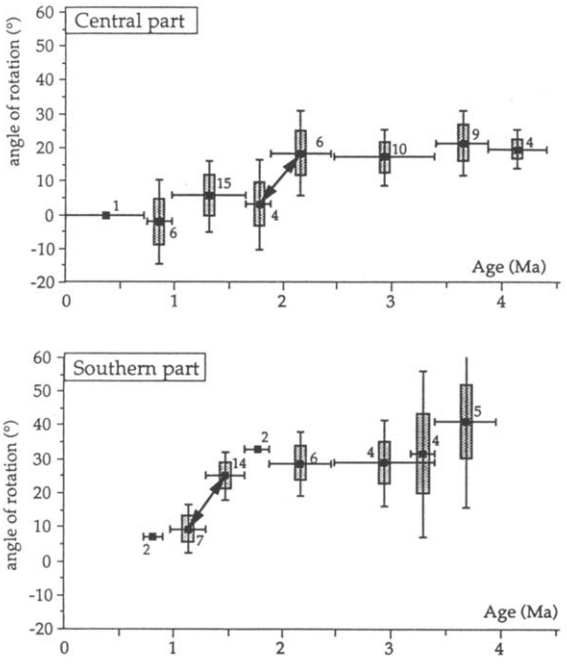


Fig. 5. Mean angles of rotation reported as a function of the age for the central and southern parts. Each point has been obtained by averaging using Fisher's statistics. The directions are corrected for the movement of the Philippine Sea plate. The number of sites on which the statistical calculation is based is shown near the point. Vertical errors correspond to α_{95} (bar) and α_{63} (dotted rectangle) projected onto the horizontal plane. No statistical parameters have been calculated when a point is based on only one or two sites. The period during which the rotation is thought to occur, and which thus marks the separation between "old" and "young" sites, is shown by the arrow.

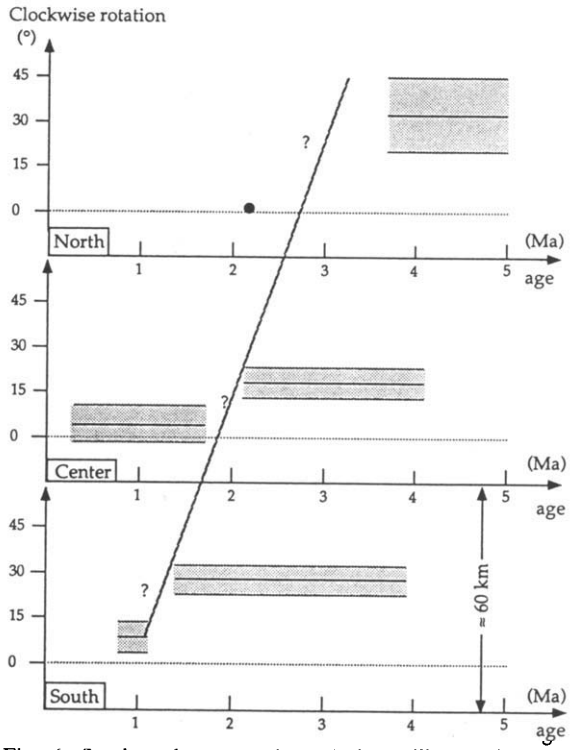


Fig. 6. Southward propagation of the collision: the mean angles of rotation for "old" and "young" sites for each region are reported as a function of the age. The width of the error zone has been calculated by projection of α_{95} (shown on the right-hand side of Fig. 4) on the horizontal plane.

which illustrates how the rotation has propagated southward (assuming that the propagation has occurred at constant speed). The slope of the line corresponds to a mean southward propagation rate of about 70 ± 10 km/Ma. It is remarkable that this value is in very good agreement with the known kinematics of the collision [10–12]. Finally, when the folds are rotated back to their original orientation, the chain appears to be aligned N–S.

These data allow a precise reconstruction of the tectonic evolution of the Coastal Range during the Plio-Pleistocene in terms of a very simple model schematized in Fig. 7. During the Pliocene period, prior to the collision and the rotation, the Coastal Range was aligned N–S and submitted to an E–W compression (Fig. 7a). This structure still exists with the same orientation south of Taiwan as observed from seismic profiles [27] and bathymetric maps. This ridge has been called "Proto-Coastal Range". In the late Pliocene, the collision started in the northern part of the Coastal Range and

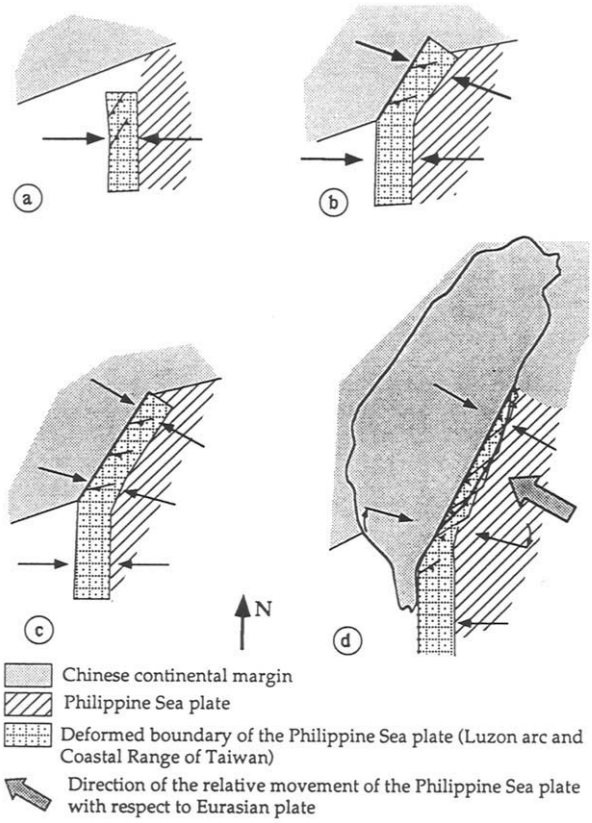


Fig. 7. Spatial and temporal evolution of the Coastal Range of Taiwan over the last few million years, before the collision occurred (a) to the present-day situation (d). The arrows illustrate the direction of the compression and its temporal and spatial evolution [19].

caused the clockwise rotation of the structures (Fig. 7b). The direction of the main compression most probably underwent a similar clockwise rotation [15]. Then, the collision and consequently the rotation migrated southward affecting successively the central and the southern part of the Coastal Range (Fig. 7c and d). The abrupt transition from the present-day orientation of the Coastal Range and the orientation of the proto-Coastal Range is consistent with a very rapid rotation as suggested by the paleomagnetic results.

6. Conclusion

This paleomagnetic study of the Coastal Range reveals some important features for the tectonic evolution of the margin of the Philippine Sea plate. The major result is that a very rapid clockwise rotation of about 25–30° has occurred in the Coastal Range since the late Pliocene. This rotation is a diachronic phenomenon with a southward propagation velocity of about 70 ± 10km/Ma, similar to the propagation rate of the collision itself.

An important observation is that the angle of rotation does not vary with the trend of the main structures. In addition, in the same structure, thrusting and folding affect not only the rotated sites but also the sites which have not recorded any significant rotation indicating that these deformations post-date, at least partly, the rotation. This shows that while the deformation of the upper cover is a continuous phenomenon, the observed rotations are on the contrary rather rapid and occurred on a time span shorter than the total deformation. Thus, the rotation appears to a large extent unrelated to the deformations (thrusts and folds) leading to the construction of the Coastal Range orogenic belt. These two points suggest that, in the collision zone, the observed rotation reflects the deformation of the Philippine Sea plate at depth, whereas folds and thrusts rather characterize the shallow deformations of the Luzon arc sedimentary sequences.

As shown by Fig. 7, the present-day configuration of the structures from south to north of the Coastal Range, i.e. from the not yet collided to the collided Luzon arc, reflects their temporal evolution from early Pliocene to late Pleistocene.

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